EEE-321

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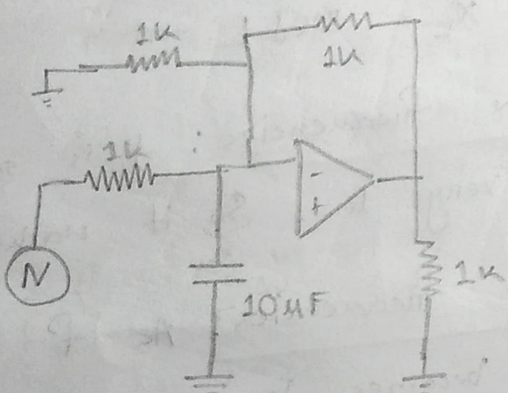
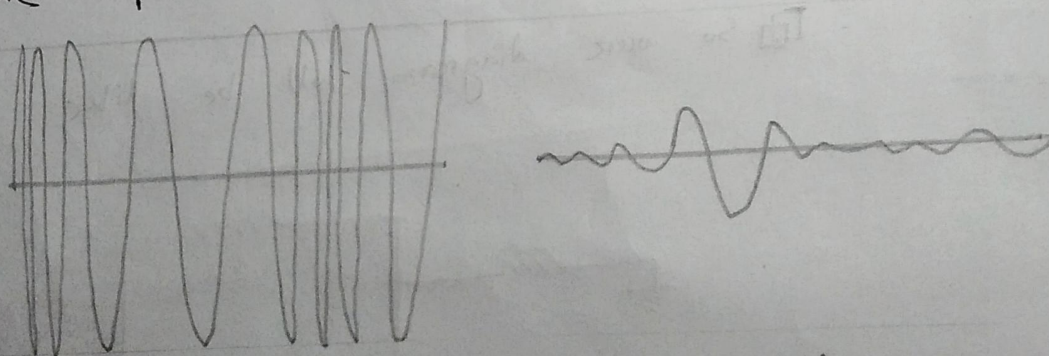
* Is it possible to make low pass filter using inductor?

Ans: In general we can make a low pass filter with following configuration:



The output will be like,



Input

Output

Here, $A_{cl} = \dfrac{V_o}{E_i} = \dfrac{1}{1 + j\omega RC}$

So, when $\omega \approx 0$, then $\dfrac{V_o}{E_i}$ is nearly 1

At high frequency $(\omega \to \infty)$, and the $\dfrac{V_o}{E_i}$ becomes 0.

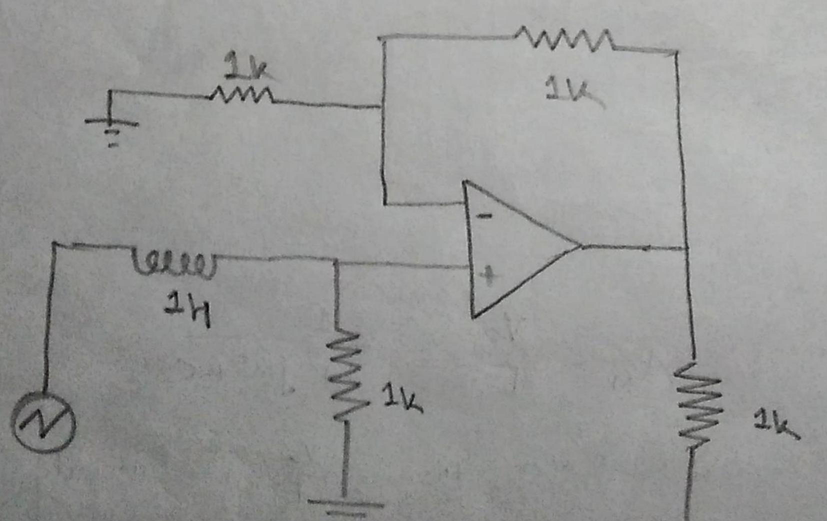
So, now if we want to use indutor instea of a conductor, we may need to put it in serie with the input. Because the resistance of an inductor (X_L) is directly proportional to the frequency.

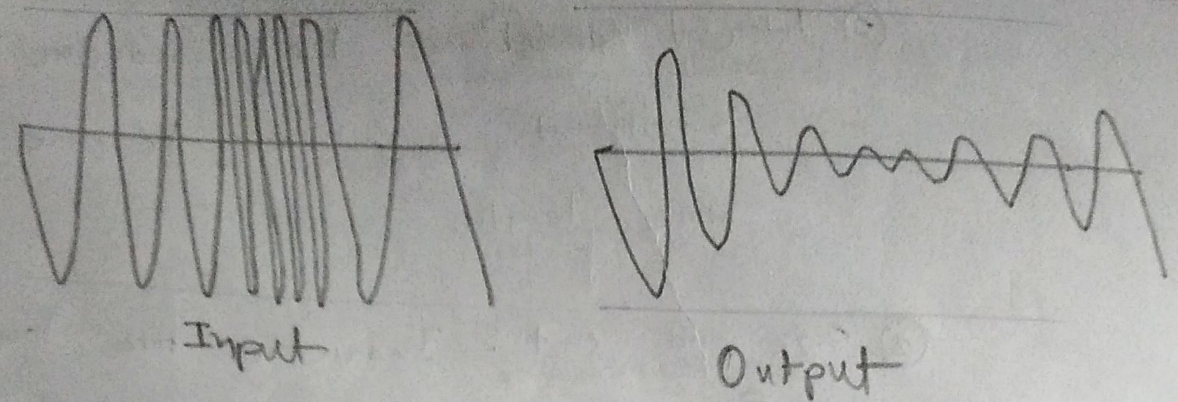
$$X_L = 2\pi f L$$

① <u>At low frequencies</u>: f is small, so X_L is very low. So it works like short circuit.

② <u>At high frequencies</u>: As f increases, X_L becomes large, which works like open circuit

So our diagram will be like,





The transfer function $\frac{V_{out}}{V_{in}} = \frac{R}{R + j\omega L}$

and the magnitude $\left|\frac{V_{out}}{V_{in}}\right| = \frac{R}{\sqrt{R^2 + (\omega L)^2}}$

At low frequency, $f = 0$, $\omega = 0$

So, $\left|\frac{V_{out}}{V_{in}}\right| = \frac{R}{\sqrt{R^2 + 0}} = 1$

So it will pass.

At high frequency, $f = \infty$ or, $\omega = \infty$

So, $\left|\frac{V_{out}}{V_{in}}\right| = \frac{R}{\sqrt{(\omega L)^2}} = 0$

So it will be blocked.

☐ So it is possible to make low pass filter with inductor. But it not practical. Here's some reasons:

1. Internal resistance: Every inductor has significant resistance because of wire length.

2. Size and cost: Inductors for low frequencies (like 100 mH or 1 H) are physically huge, heavy and expensive.

3. Magnatic interference: Inductors are prone to electromagnetic interference (EMI). And they are also hard to implement in a silicon chip.

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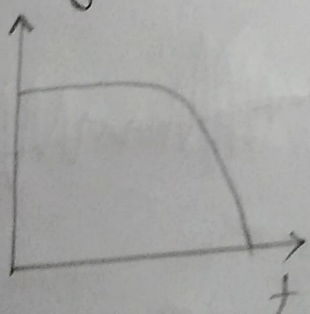
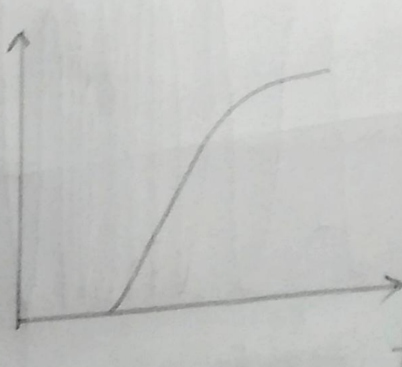
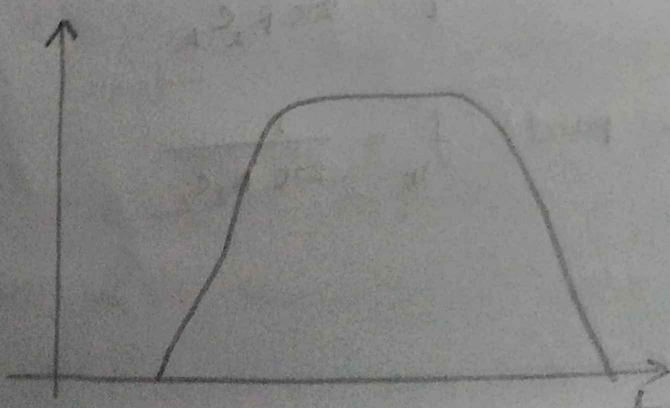
① Bandpass filter implementation.

Ans: A low pass filter pass filter passes frequencies that are below the cutoff frequency and a high pass filter passes frequencies that are above the cutoff frequency.

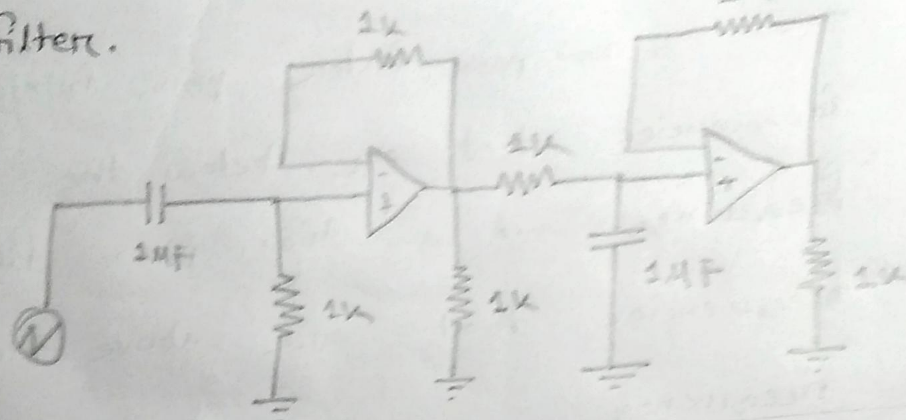


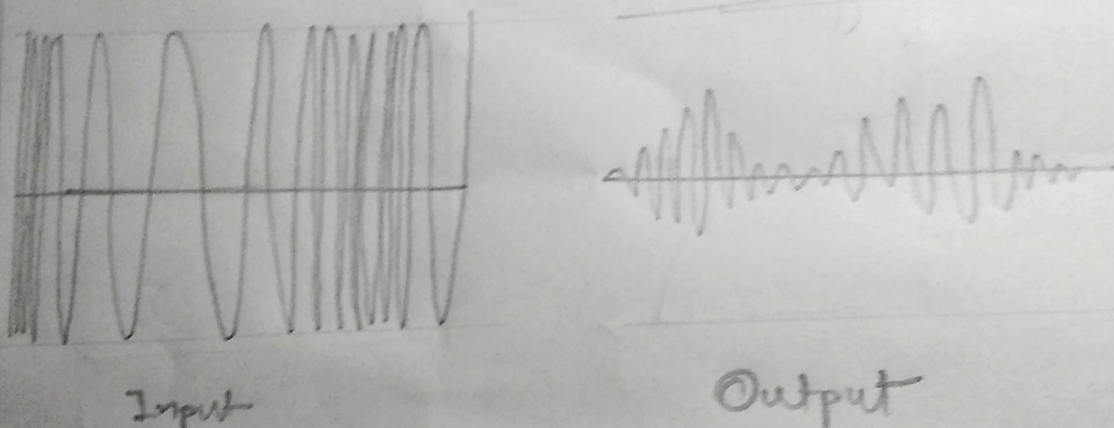


And in the band pass, it passess frequencies that fall only within a relatively narrow range.



So, we can combine both high pass filter and low pass filter to make a band-pass filter.





So, the cutoff region for

lower part $f_L = \frac{1}{2\pi R_1 C_1}$

upper part $f_H = \frac{1}{2\pi R_2 C_2}$